

UNIT-2: CORRELATION, REGRESSION, AND TIME SERIES FORECASTING

An Exhaustive, Self-Contained Academic Reference Manual with Deep Theoretical Derivations, Comprehensive Numerical Layouts, and 30+ Extensively Worked Step-by-Step Practical Application Problems.

Subject Coverage:

Linear & Non-Linear Correlation • Karl Pearson's Product-Moment • Spearman's Rank Coefficient • Concurrent Deviations • Linear Regression Modeling • Ordinary Least Squares (OLS) Equations • Standard Error of Estimate • Business Forecasting Methodologies • Quantitative & Qualitative Frameworks • Decomposition of Time Series • Trend Fitting via Moving Averages & Least Squares.

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MODULE 1: CORRELATION ANALYSIS

1.1 Conceptual Foundations and Bivariate Distributions

In statistical investigations, a univariate distribution involves analyzing a single variable isolated from external interactions (e.g., studying the heights of employees within an enterprise). However, economic and operational realities demand the study of **bivariate distributions**, where two distinct characteristics are measured simultaneously for each sampling unit. Correlation analysis is the branch of statistical science concerned with quantifying the existence, direction, and magnitude of a mutual relationship between two or more variables.

When two variables vary in such a manner that changes in the value of one variable are accompanied by systematic changes in the value of the other, the variables are said to be **correlated**. In a strict cause-and-effect structure, an independent variable directly modifies a dependent variable. However, correlation does not inherently imply causation. A strong statistical association may arise from:

1. *Direct causal dependence*: Variable A directly drives Variable B (e.g., Advertising Expenditures driving Total Sales Revenue).
2. *Common external drivers*: Both variables are governed by a third, unobserved factor (e.g., both agricultural yield and consumer discretionary spending rising due to a favorable monsoon season).
3. *Mutual systemic feedback*: The variables influence each other in an ongoing loop (e.g., Price and Demand dynamics).
4. *Spurious or accidental alignment*: A statistically high correlation occurring entirely by chance without any structural or logical connection (e.g., the import of chocolate matching the birth rate in an unrelated region).

Classification of Correlation

Correlation configurations can be systematically grouped across three primary diagnostic dimensions:

1. Directionality: Positive vs. Negative Correlation

- *Positive (Direct) Correlation*: Occurs when variables move in the same direction. An increase in the independent variable matches an increase in the dependent variable, or a decrease matches a decrease. Examples: Income and Expenditures; Temperature and Ice Cream Sales; Engine Size and Fuel Consumption.
- *Negative (Inverse) Correlation*: Occurs when variables move in diametrically opposed directions. An increase in one variable corresponds to a systematic decrease in the other. Examples: Commodity Supply and Market Price; Unemployment Rate and Consumer Confidence Index; Automotive Speed and Travel Time.

2. Structural Linearity: Linear vs. Non-Linear (Curvilinear) Correlation

- *Linear Correlation*: Present when the ratio of change between the two variables remains perfectly constant across their entire distribution. If plotted on a Cartesian coordinate plane, the data points arrange themselves along a straight line. Formally, it satisfies the linear equation: $Y = a + bX$.
- *Non-Linear (Curvilinear) Correlation*: Present when the ratio of change varies across different intervals of the variables. The relationship changes character as the values scale, forming a curve on a Cartesian plot. Examples include diminishing marginal returns or U-shaped average cost curves. Formally modeled via polynomials or exponential equations: $Y = a + bX + cX^2$.

3. Dimensionality: Simple, Multiple, and Partial Correlation

- *Simple Correlation*: Restricts the structural investigation to exactly two variables (e.g., Rainfall and Crop Yield).
- *Multiple Correlation*: Studies the joint relationship among three or more variables simultaneously (e.g., examining how Crop Yield changes based on Rainfall, Fertilizer application volume, and Ambient Temperature combined).
- *Partial Correlation*: Examines the relationship between two variables while analytically neutralizing or holding constant the confounding effects of one or more additional variables. For instance, analyzing the precise relationship between Yield and Fertilizer while mathematically stripping away the variability caused by fluctuating Rainfall.

1.2 Geometric Interpretation: Scatter Diagrams

The **Scatter Diagram** (or Scatter Plot) represents the foundational visual methodology for exploring bivariate data. Each pair of observations (x_i, y_i) is plotted as a single localized point on a two-dimensional Cartesian grid, where the horizontal X axis represents the independent/predictor variable and the vertical Y axis represents the dependent/response variable.

The visual configuration formed by the cloud of plotted points reveals the underlying pattern of the relationship:

- **Perfect Positive Correlation**: All points fall precisely on a single straight line sloping upward from the bottom-left quadrant to the top-right quadrant. The correlation coefficient is exactly $r = +1$.
- **Highly Positive Correlation**: The points do not lie precisely on a single line but form a tight, narrow band trending upward. This indicates a strong positive association.
- **Perfect Negative Correlation**: All points fall precisely on a single straight line sloping downward from the top-left quadrant to the bottom-right quadrant. The correlation coefficient is exactly $r = -1$.
- **Highly Negative Correlation**: The points form a tight, narrow band trending downward, indicating a strong inverse relationship.
- **Zero / Absence of Correlation**: The points are distributed across the grid in an unorganized cloud, circle, or square pattern with no discernible linear path or slope. This indicates that changes in X provide zero predictable information about Y . The correlation coefficient is $r = 0$.



Perfect Positive (r = +1)



Perfect Negative (r = -1)

While scatter diagrams are useful for initial exploratory data analysis, they are purely qualitative and subjective. They indicate the presence and general direction of a relationship but cannot provide an exact numerical measure of its strength. To obtain this precise measurement, we must compute numerical correlation coefficients.

1.3 Karl Pearson's Product-Moment Coefficient

Developed by the English mathematician Karl Pearson, the **Product-Moment Correlation Coefficient** (denoted by r) is the standard mathematical metric used to measure the strength and direction of a linear relationship between two ratio or interval-scaled variables. It is defined as the ratio of the covariance of the two variables to the product of their respective standard deviations.

Mathematical Formulations

1. The Fundamental Covariance Formula (Conceptual Formula)

The standard definition of the Pearson correlation coefficient is:

$$r = \text{Cov}(X, Y) / [\sigma_x \times \sigma_y]$$

Where the covariance between X and Y is defined as:

$$\text{Cov}(X, Y) = [\Sigma(X - \bar{X})(Y - \bar{Y})] / n$$

And the standard deviations σ_x and σ_y are defined as:

$$\sigma_x = \sqrt{[\Sigma(X - \bar{X})^2 / n]} \quad \sigma_y = \sqrt{[\Sigma(Y - \bar{Y})^2 / n]}$$

2. The Product-Moment Direct Formula (Using Actual Means)

By substituting the expanded terms into the foundational ratio, the sample size parameter n cancels out, yielding the operational direct formula:

$$r = \Sigma(X - \bar{X})(Y - \bar{Y}) / \sqrt{[\Sigma(X - \bar{X})^2 \times \Sigma(Y - \bar{Y})^2]}$$

Letting lowercase deviations be defined as $x = (X - \bar{X})$ and $y = (Y - \bar{Y})$, the expression can be compactly written as:

$$r = \Sigma xy / \sqrt{[\Sigma x^2 \times \Sigma y^2]}$$

3. The Raw Data Direct Calculation Formula

When working with raw data where the arithmetic means $\bar{X}$ and $\bar{Y}$ are fractional numbers, using mean-deviation equations becomes tedious and prone to rounding errors. In these cases, it is much more efficient to use the raw data direct formula:

$$r = [n \times \Sigma XY - (\Sigma X)(\Sigma Y)] / \sqrt{[n \times \Sigma X^2 - (\Sigma X)^2] \times [n \times \Sigma Y^2 - (\Sigma Y)^2]}$$

4. The Short-cut Formula (Using Assumed Means)

To simplify calculations when managing large, un-grouped raw numbers without a computer, you can choose arbitrary values as **assumed means** (denoted A_x and A_y). Define the operational deviations as $d_x = X - A_x$ and $d_y = Y - A_y$. The formula then becomes:

$$r = [n \times \Sigma d_x d_y - (\Sigma d_x)(\Sigma d_y)] / \sqrt{[n \times \Sigma d_x^2 - (\Sigma d_x)^2] \times [n \times \Sigma d_y^2 - (\Sigma d_y)^2]}$$

5. The Step-Deviation Formula

When the data values have a constant class interval or common factor (denoted C_x and C_y), you can scale down the numbers further using step-deviations: $d'_x = (X - A_x) / C_x$ and $d'_y = (Y - A_y) / C_y$. The formula remains structurally identical because the scale scaling constants cancel out completely:

$$r = [n \times \Sigma d'_x d'_y - (\Sigma d'_x)(\Sigma d'_y)] / \sqrt{[n \times \Sigma d'^2_x - (\Sigma d'_x)^2] \times [n \times \Sigma d'^2_y - (\Sigma d'_y)^2]}$$

1.4 Assumptions, Limitations, and Analytical Properties of r

Underlying Assumptions

For Karl Pearson's coefficient to provide valid analytical conclusions, the bivariate dataset must satisfy several strict statistical assumptions:

1. **Linearity of Relationships:** The relationship between the two variables must be linear. If the true relationship is curvilinear, Pearson's r will underestimate the strength of the association, often yielding a value close to zero despite a strong non-linear pattern.
2. **Continuous Scaled Variables:** Both variables must be quantitative and measured on a continuous interval or ratio scale (e.g., height, revenue, temperature, time). It cannot be directly applied to qualitative or nominal data.
3. **Bivariate Normal Distribution:** The joint distribution of the two variables must follow a bivariate normal distribution. This means that for any specific value of X , the corresponding values of Y must be normally distributed, and vice versa.
4. **Homoscedasticity:** The variance of the data points around the line of best fit must be roughly constant across all values of the independent variable. If the spread fans out or narrows significantly (heteroscedasticity), the reliability of the coefficient decreases.
5. **Absence of Outliers:** Pearson's coefficient is highly sensitive to extreme outliers. A single outlier can artificially inflate or deflate the calculated value of r , leading to incorrect interpretations.

Fundamental Properties of r (With Proofs)

Property 1: The Bounds of Correlation. The Karl Pearson correlation coefficient is strictly bounded between the values of -1 and +1. Formally:

$$-1 \leq r \leq +1$$

Analytical Proof: Let $x = X - \bar{X}$ and $y = Y - \bar{Y}$. Consider two standardized variables $z_x = x / \sigma_x$ and $z_y = y / \sigma_y$. We know that the sum of squared numbers must always be non-negative. Therefore, if we sum the squared differences between these standardized scores, the result must be greater than or equal to zero:

$$\Sigma(z_x - z_y)^2 \geq 0 \Rightarrow \Sigma(z_x^2 + z_y^2 - 2z_x z_y) \geq 0$$

$$\Sigma z_x^2 + \Sigma z_y^2 - 2\Sigma z_x z_y \geq 0$$

By definition, $\sum z_x^2 = \sum (x/\sigma_x)^2 = \sum x^2/\sigma_x^2 = n$ (since $\sigma_x^2 = \sum x^2/n$). Similarly, $\sum z_y^2 = n$. Furthermore, the term $\sum z_x z_y = \sum (xy / (\sigma_x \sigma_y)) = (1 / (\sigma_x \sigma_y)) \times \sum xy$. Since $r = \sum xy / (n \sigma_x \sigma_y)$, we have $\sum z_x z_y = n \cdot r$. Substituting these back into the inequality gives:

$$n + n - 2(n \cdot r) \geq 0 \Rightarrow 2n - 2nr \geq 0 \Rightarrow 2n(1 - r) \geq 0$$

Since the sample size n is always positive, we can divide both sides by $2n$, yielding:

$$1 - r \geq 0 \Rightarrow r \leq 1$$

Next, we apply the same logic to the sum of squared additions, which must also be non-negative:

$$\sum (z_x + z_y)^2 \geq 0 \Rightarrow \sum z_x^2 + \sum z_y^2 + 2\sum z_x z_y \geq 0$$

$$n + n + 2nr \geq 0 \Rightarrow 2n(1 + r) \geq 0 \Rightarrow 1 + r \geq 0 \Rightarrow r \geq -1$$

Combining the two inequalities ($r \leq 1$ and $r \geq -1$) proves that the correlation coefficient must always fall within the range $[-1, +1]$.

Property 2: Invariance to Origin and Scale Changes. The correlation coefficient r is completely unaffected by changes to the origin or the scale of the underlying data.

This means that if you shift the data by adding/subtracting a constant (origin shift) or multiply/divide the data by a factor (scale shift), the resulting correlation coefficient remains unchanged. Formally, if we define new variables $u = (X - a) / h$ and $v = (Y - b) / k$, where a, b, h, k are constants and $h > 0, k > 0$, then:

$$r_{xy} = r_{uv}$$

This property is highly useful because it allows us to simplify complex calculations by scaling large numbers down into smaller, more manageable values using the step-deviation method.

Interpretation Framework for Pearson's Coefficient

Numerical Value of Coefficient (r)	Qualitative Classification of Relationship Strength
$r = +1.00$	Perfect Positive Linear Correlation
$+0.75 \leq r < +1.00$	Very High / Strong Positive Linear Correlation
$+0.50 \leq r < +0.75$	Moderate Positive Linear Correlation
$+0.25 \leq r < +0.50$	Low / Weak Positive Linear Correlation
$0.00 < r < +0.25$	Negligible Positive Linear Association
$r = 0.00$	Complete Absence of Linear Correlation
$-0.25 < r < 0.00$	Negligible Negative Linear Association
$-0.50 < r \leq -0.25$	Low / Weak Negative Linear Correlation
$-0.75 < r \leq -0.50$	Moderate Negative Linear Correlation
$-1.00 < r \leq -0.75$	Very High / Strong Negative Linear Correlation
$r = -1.00$	Perfect Negative Linear Correlation

1.5 Probable Error and Significance Testing

The **Probable Error** of the correlation coefficient (denoted $P.E._r$) is a statistical measure used to evaluate the reliability and significance of the calculated sample correlation coefficient r relative to the broader population. It helps determine whether a calculated correlation represents a true relationship or is simply a result of sampling error.

The mathematical formula for Probable Error is derived directly from the standard error of a normal distribution:

$$P.E._r = 0.6745 \times [(1 - r^2) / \sqrt{n}]$$

Where r is the calculated sample correlation coefficient and n is the total number of paired observations. The constant 0.6745 comes from the fact that in a standard normal distribution, 50% of the values fall within ± 0.6745 standard deviations of the mean.

The underlying **Standard Error** ($S.E._r$) of the correlation coefficient is defined as:

$$S.E._r = (1 - r^2) / \sqrt{n}$$

Decision Criteria for Statistical Significance

To evaluate the significance of a calculated correlation coefficient using Probable Error, we apply the following guidelines:

1. If the absolute value of the correlation coefficient is less than its Probable Error ($|r| < P.E._r$), the correlation is considered statistically insignificant. There is no evidence of a real relationship; the observed correlation is likely due to random sampling variation.
2. If the absolute value of the correlation coefficient is more than six times its Probable Error ($|r| > 6 \times P.E._r$), the correlation is considered highly significant. It is extremely likely that a real linear relationship exists between the variables in the population.
3. The population correlation coefficient can be estimated to fall within the following confidence limits: $r \pm P.E._r$.

1.6 Spearman's Rank Correlation Coefficient

When analyzing real-world data, we often encounter scenarios where characteristics cannot be measured precisely on a continuous quantitative scale. Instead, they are qualitative attributes that can only be ranked in order of preference, capability, or performance (e.g., assessing leadership ability, artistic beauty, intelligence, or brand preference). In these qualitative contexts, Karl Pearson's product-moment method cannot be used.

To solve this, Charles Spearman introduced the **Rank Correlation Coefficient** (denoted by **R** or **ρ**). This non-parametric technique measures the linear relationship between two sets of ranked data. It works by calculating the differences between the ranks assigned to each observation.

Mathematical Formulations for Spearman's Rank Coefficient

Case A: When Ranks are Distinct and No Ties Exist

If every item in the dataset has a unique rank with no duplicates, the formula for Spearman's rank correlation coefficient is:

$$R = 1 - [(6 \times \sum D^2) / (n \times (n^2 - 1))]$$

Where:

$D = R_1 - R_2$ represents the difference between the ranks assigned to an item across the two variables.

$\sum D^2$ is the sum of these squared rank differences.

n is the total number of paired observations in the sample.

Case B: When Ranks are Tied (Repeated Ranks)

In practice, two or more observations may share identical values, resulting in a tie. When this happens, each tied observation is assigned the **average (mean) rank** of the positions they would have occupied if they were distinct.

Because ties reduce the overall variance of the ranks, we must apply a correction factor to the formula. For each group of tied values, we calculate a correction term: $m(m^2 - 1) / 12$, where m represents the number of observations tied at that specific rank. These correction terms are then added to ΣD^2 in the numerator:

$$R = 1 - [6 \times \{ \Sigma D^2 + [m_1(m_1^2 - 1)/12] + [m_2(m_2^2 - 1)/12] + \dots \}] / [n \times (n^2 - 1)]$$

Where m_1 is the number of items tied in the first group, m_2 is the number tied in the second group, and so on. This adjusted formula ensures that the correlation coefficient remains accurate even when multiple ties are present in the dataset.

1.7 Comprehensive Bivariate Problem Sets on Correlation

PROBLEM 1: KARL PEARSON DIRECT CALCULATION METHOD (ACTUAL MEANS)

A corporate analytics department wants to evaluate the relationship between monthly promotional spend and regional sales volume. The following data shows the expenditures (X, in thousands of dollars) and sales volumes (Y, in hundreds of units) across 6 regional territories:

Promotional Spend (X): 10, 12, 15, 18, 20, 21

Sales Volume (Y): 40, 42, 50, 54, 60, 62

Calculate Karl Pearson's correlation coefficient using the actual mean method and interpret the result.

SOLUTION

First, we calculate the arithmetic means for both variables:

$$n = 6$$

$$\Sigma X = 10 + 12 + 15 + 18 + 20 + 21 = 96 \Rightarrow \bar{X} = 96 / 6 = 16$$

$$\Sigma Y = 40 + 42 + 50 + 54 + 60 + 62 = 308 \Rightarrow \bar{Y} = 308 / 6 = 51.333$$

Since the mean of Y is fractional, let's construct the structural computation matrix using actual deviations to see how the numbers line up, but to ensure high accuracy we can also verify via the raw data layout. Let's use actual deviations: $x = X - 16$ and $y = Y - 51.333$.

X	Y	$x = X - 16$	$y = Y - 51.333$	x^2	y^2	xy
10	40	-6	-11.333	36	128.437	68.00
12	42	-4	-9.333	16	87.107	37.33
15	50	-1	-1.333	1	1.777	1.33
18	54	2	2.667	4	7.113	5.33
20	60	4	8.667	16	75.117	34.67
21	62	5	10.667	25	113.785	53.33
$\Sigma=96$	$\Sigma=308$	$\Sigma=0$	$\Sigma=0$	$\Sigma x^2=98$	$\Sigma y^2=413.336$	$\Sigma xy=200.00$

Now, substitute these sums into Karl Pearson's direct formula:

$$r = \Sigma xy / \sqrt{[\Sigma x^2 \times \Sigma y^2]} = 200.00 / \sqrt{[98 \times 413.336]}]$$

$$r = 200.00 / \sqrt{[40506.928]} = 200.00 / 201.263 = 0.9937$$

Analytical Interpretation: The calculated correlation coefficient is $r = +0.9937$. This indicates an extremely strong, near-perfect positive linear relationship between promotional spend and sales volume. Increases in promotional expenditure are highly aligned with predictable increases in regional sales.

PROBLEM 2: RAW DATA FORMULA APPLICATION (FRACTIONAL MEANS)

A quality control engineer monitors the relationship between curing oven temperature (X, in degrees Celsius) and the structural tensile strength of an alloy component (Y, in MPa). Calculate the correlation coefficient directly from the following raw values:

Oven Temp (X): 105, 110, 112, 118, 125

Strength (Y): 320, 325, 330, 345, 360

SOLUTION

We use the raw data direct formula: $r = [n\Sigma XY - (\Sigma X)(\Sigma Y)] / \sqrt{[n\Sigma X^2 - (\Sigma X)^2] \times [n\Sigma Y^2 - (\Sigma Y)^2]}$ with $n = 5$.

X	Y	X ²	Y ²	XY
105	320	11,025	102,400	33,600
110	325	12,100	105,625	35,750
112	330	12,544	108,900	36,960
118	345	13,924	119,025	40,710
125	360	15,625	129,600	45,000
Σ=564	Σ=1680	ΣX²=65,218	ΣY²=565,550	ΣXY=192,020

Substitute the values into the formula:

$$\text{Numerator} = 5(192,020) - (564)(1680) = 960,100 - 947,520 = 12,580$$

$$\text{Denominator X-term} = 5(65,218) - (564)^2 = 326,090 - 318,096 = 7,994$$

$$\text{Denominator Y-term} = 5(565,550) - (1680)^2 = 2,827,750 - 2,822,400 = 5,350$$

$$r = 12,580 / \sqrt{[7,994 \times 5,350]} = 12,580 / \sqrt{[42,767,900]} = 12,580 / 6,539.717 = 0.9924$$

Analytical Interpretation: The coefficient $r = +0.9924$ confirms an exceptionally strong, positive linear association between curing oven temperature and component tensile strength.

PROBLEM 3: SHORT-CUT METHOD VIA ASSUMED MEAN FRAMEWORK

The table below shows indices for raw material prices (X) and finished wholesale retail items (Y) over seven consecutive months. Using the short-cut method with assumed means, calculate the Pearson coefficient.

Material Price (X): 65, 66, 67, 67, 68, 69, 71

Retail Items (Y): 120, 123, 125, 126, 130, 131, 135

SOLUTION

Let the assumed mean for X be $A_x = 67$ and the assumed mean for Y be $A_y = 126$. We compute the deviations: $d_x = X - 67$ and $d_y = Y - 126$ with $n = 7$.

X	Y	d_x	d_y	d_x^2	d_y^2	$d_x d_y$
65	120	-2	-6	4	36	12
66	123	-1	-3	1	9	3
67	125	0	-1	0	1	0
67	126	0	0	0	0	0
68	130	1	4	1	16	4
69	131	2	5	4	25	10
71	135	4	9	16	81	36
-	-	$\Sigma d_x = 4$	$\Sigma d_y = 8$	$\Sigma d_x^2 = 26$	$\Sigma d_y^2 = 168$	$\Sigma d_x d_y = 65$

Substitute these values into the short-cut formula:

$$\text{Numerator} = 7(65) - (4)(8) = 455 - 32 = 423$$

$$\text{Denominator X-term} = 7(26) - (4)^2 = 182 - 16 = 166$$

$$\text{Denominator Y-term} = 7(168) - (8)^2 = 1176 - 64 = 1112$$

$$r = 423 / \sqrt{[166 \times 1112]} = 423 / \sqrt{184592} = 423 / 429.641 = 0.9845$$

Interpretation: A very strong positive linear correlation ($r = 0.9845$) exists, showing a clear pass-through effect from raw materials to finished wholesale goods.

PROBLEM 4: SIGNIFICANCE VERIFICATION USING PROBABLE ERROR

Using the correlation coefficient calculated in Problem 3 ($r = 0.9845$, $n = 7$), determine the Probable Error and evaluate whether the relationship is statistically significant.

SOLUTION

The formula for Probable Error is: $P.E._r = 0.6745 \times [(1 - r^2) / \sqrt{n}]$

First, calculate r^2 : $0.9845^2 = 0.96924$

Then, find $1 - r^2$: $1 - 0.96924 = 0.03076$

The square root of n is: $\sqrt{7} = 2.64575$

Now, substitute these values into the formula:

$$P.E._r = 0.6745 \times [0.03076 / 2.64575] = 0.6745 \times 0.011626 = 0.00784$$

To evaluate statistical significance, we multiply the Probable Error by 6:

$$6 \times P.E._r = 6 \times 0.00784 = 0.04704$$

Comparing our correlation coefficient ($|r| = 0.9845$) to $6 \times P.E._r$:

$$0.9845 > 0.04704 \Rightarrow |r| > 6 \times P.E._r$$

Conclusion: Because the calculated correlation coefficient is substantially greater than six times its Probable Error, the relationship is highly statistically significant. We can be confident that this positive correlation reflects a true pattern in the broader population rather than a random sampling artifact.

PROBLEM 5: SPEARMAN'S RANK CORRELATION (DISTINCT RANKS, NO TIES)

Two investment analysts grade 8 high-yield technology corporate bonds based on credit stability metrics. Their assignments are given as follows:

Bond ID: A, B, C, D, E, F, G, H

Analyst 1 Rank (R1): 4, 1, 5, 2, 6, 7, 3, 8

Analyst 2 Rank (R2): 5, 2, 4, 1, 7, 8, 3, 6

Calculate Spearman's rank correlation coefficient to assess the level of agreement between the two analysts.

SOLUTION

Since these are distinct ranks with no ties, we can use the standard Spearman formula: $R = 1 - [(6 \times \Sigma D^2) / (n \times (n^2 - 1))]$, where $n = 8$.

Bond	Rank R1	Rank R2	D = R1 - R2	D ²
A	4	5	-1	1
B	1	2	-1	1
C	5	4	1	1
D	2	1	1	1
E	6	7	-1	1
F	7	8	-1	1
G	3	3	0	0
H	8	6	2	4
Total	-	-	$\Sigma D = 0$	$\Sigma D^2 = 10$

Substitute these values into the formula:

$$R = 1 - [(6 \times 10) / (8 \times (8^2 - 1))] = 1 - [60 / (8 \times 63)]$$

$$R = 1 - [60 / 504] = 1 - 0.1190 = 0.8810$$

Interpretation: The rank correlation coefficient is $R = 0.8810$. This indicates a strong positive consensus between the two analysts regarding the credit stability of the corporate bonds.

PROBLEM 6: SPEARMAN'S RANK CORRELATION WITH TIED OBSERVATIONS

Calculate the rank correlation coefficient for the following raw scores achieved by 8 students in financial economics (X) and business analytics (Y):

Economics (X): 80, 75, 75, 60, 50, 80, 75, 40

Analytics (Y): 90, 85, 90, 70, 60, 85, 80, 50

SOLUTION

First, we need to assign ranks to the raw scores. Since several values are repeated, we will use the average rank method for tied values.

Ranking Variable X:

The highest score is 80, which appears twice (occupying positions 1 and 2). Average rank = $(1 + 2) / 2 = 1.5$.

The next score is 75, which appears three times (occupying positions 3, 4, and 5). Average rank = $(3 + 4 + 5) / 3 = 4$.

The remaining scores are unique: 60 gets rank 6, 50 gets rank 7, and 40 gets rank 8.

Our tied groups for X are: $m_1 = 2$ (for 80) and $m_2 = 3$ (for 75).

Ranking Variable Y:

The highest score is 90, which appears twice (occupying positions 1 and 2). Average rank = $(1 + 2) / 2 = 1.5$.

The next score is 85, which appears twice (occupying positions 3 and 4). Average rank = $(3 + 4) / 2 = 3.5$.

The remaining scores are unique: 80 gets rank 5, 70 gets rank 6, 60 gets rank 7, and 50 gets rank 8.

Our tied groups for Y are: $m_3 = 2$ (for 90) and $m_4 = 2$ (for 85).

X	Y	Rank R1	Rank R2	D = R1 - R2	D ²
80	90	1.5	1.5	0.0	0.00
75	85	4.0	3.5	0.5	0.25
75	90	4.0	1.5	2.5	6.25
60	70	6.0	6.0	0.0	0.00
50	60	7.0	7.0	0.0	0.00
80	85	1.5	3.5	-2.0	4.00
75	80	4.0	5.0	-1.0	1.00
40	50	8.0	8.0	0.0	0.00
Total	-	-	-	$\Sigma D = 0$	$\Sigma D^2 = 11.5$

Now, calculate the correction factors for each tied group:

For $m_1 = 2$: $2(2^2 - 1) / 12 = 6 / 12 = 0.5$

For $m_2 = 3$: $3(3^2 - 1) / 12 = 24 / 12 = 2.0$

For $m_3 = 2$: $2(2^2 - 1) / 12 = 6 / 12 = 0.5$

For $m_4 = 2$: $2(2^2 - 1) / 12 = 6 / 12 = 0.5$

Total Correction = $0.5 + 2.0 + 0.5 + 0.5 = 3.5$

Add the correction to ΣD^2 and solve using the adjusted formula with $n = 8$:

$$\text{Adjusted } \Sigma D^2 = 11.5 + 3.5 = 15.0$$

$$R = 1 - [(6 \times 15.0) / (8 \times (8^2 - 1))] = 1 - [90 / (8 \times 63)] = 1 - [90 / 504]$$

$$R = 1 - 0.1786 = 0.8214$$

Interpretation: The rank correlation coefficient is $R = 0.8214$. This shows a strong positive relationship between performance in financial economics and business analytics, even after correcting for multiple tied ranks.

PROBLEM 7: DIRECT COMPUTATION MATRIX AND COVARIANCE DERIVATION

The following data represents the volume of production (X, in thousands of units) and the corresponding unit production cost (Y, in dollars) at a manufacturing plant over 5 periods:

Production (X): 2, 4, 6, 8, 10

Unit Cost (Y): 25, 22, 20, 15, 13

Calculate the covariance and the Pearson correlation coefficient using the raw data layout.

SOLUTION

Let's construct our data table with $n = 5$:

X	Y	X ²	Y ²	XY
2	25	4	625	50
4	22	16	484	88
6	20	36	400	120
8	15	64	225	120
10	13	100	169	130
$\Sigma=30$	$\Sigma=95$	$\Sigma X^2=220$	$\Sigma Y^2=1903$	$\Sigma XY=508$

First, find the means:

$$\bar{X} = 30 / 5 = 6$$

$$\bar{Y} = 95 / 5 = 19$$

Next, calculate the covariance:

$$\text{Cov}(X, Y) = (\Sigma XY / n) - (X \times \bar{Y}) = (508 / 5) - (6 \times 19) = 101.6 - 114 = -12.4$$

Now, calculate the variances and standard deviations:

$$\sigma_x^2 = (\Sigma X^2 / n) - \bar{X}^2 = (220 / 5) - 6^2 = 44 - 36 = 8 \Rightarrow \sigma_x = \sqrt{8} = 2.8284$$

$$\sigma_y^2 = (\Sigma Y^2 / n) - \bar{Y}^2 = (1903 / 5) - 19^2 = 380.6 - 361 = 19.6 \Rightarrow \sigma_y = \sqrt{19.6} = 4.4272$$

Finally, solve for the correlation coefficient:

$$r = \text{Cov}(X, Y) / [\sigma_x \times \sigma_y] = -12.4 / [2.8284 \times 4.4272] = -12.4 / 12.5218 = -0.9903$$

Interpretation: The correlation coefficient is $r = -0.9903$. This reveals an exceptionally strong negative linear relationship, reflecting how economies of scale help lower unit production costs as manufacturing volume increases.

PROBLEM 8: RECONSTRUCTION OF CORE VALUES FROM SUMMATION INPUTS

An analyst loses the original sheets of a bivariate dataset but retains the following summary statistics:

$n = 10$, $\Sigma X = 140$, $\Sigma Y = 150$, $\Sigma(X - 10)^2 = 180$, $\Sigma(Y - 15)^2 = 215$, and $\Sigma(X - 10)(Y - 15) = 60$.

Reconstruct the value of Karl Pearson's correlation coefficient.

SOLUTION

Let $A_x = 10$ and $A_y = 15$ be our assumed means. This allows us to map the given values to our short-cut deviations:

$$d_x = X - 10 \Rightarrow \Sigma d_x^2 = 180$$

$$d_y = Y - 15 \Rightarrow \Sigma d_y^2 = 215$$

$$\Sigma d_x d_y = 60$$

Next, we calculate the sums of these deviations, Σd_x and Σd_y :

$$\Sigma d_x = \Sigma(X - 10) = \Sigma X - 10n = 140 - 10(10) = 140 - 100 = 40$$

$$\Sigma d_y = \Sigma(Y - 15) = \Sigma Y - 15n = 150 - 15(10) = 150 - 150 = 0$$

Now, substitute these values into the short-cut formula:

$$\text{Numerator} = n \Sigma d_x d_y - (\Sigma d_x)(\Sigma d_y) = 10(60) - (40)(0) = 600$$

$$\text{Denominator X-term} = n\Sigma d_x^2 - (\Sigma d_x)^2 = 10(180) - (40)^2 = 1800 - 1600 = 200$$

$$\text{Denominator Y-term} = n\Sigma d_y^2 - (\Sigma d_y)^2 = 10(215) - (0)^2 = 2150$$

$$r = 600 / \sqrt{[200 \times 2150]} = 600 / \sqrt{[430,000]} = 600 / 655.744 = 0.9150$$

Interpretation: The reconstructed correlation coefficient is $r = 0.9150$, confirming a strong positive linear relationship between the two variables.

PROBLEM 9: CORRECTING FOR ERRORS IN RECORDED DATA PAIRS

A statistical calculation based on 20 pairs of observations yields a correlation coefficient of $r = 0.45$. During a data audit, the team discovers that one data pair was recorded incorrectly: the values ($X = 12, Y = 15$) were entered instead of the correct values ($X = 10, Y = 12$).

Given the original summary statistics: $\Sigma X = 200$, $\Sigma Y = 240$, $\Sigma X^2 = 2220$, $\Sigma Y^2 = 3010$, and $\Sigma XY = 2460$, calculate the corrected correlation coefficient.

SOLUTION

To fix this, we subtract the incorrect values and add the correct values for each summary statistic ($n = 20$):

$$\text{Corrected } \Sigma X: 200 - 12 + 10 = 198$$

$$\text{Corrected } \Sigma Y: 240 - 15 + 12 = 237$$

$$\text{Corrected } \Sigma X^2: 2220 - 12^2 + 10^2 = 2220 - 144 + 100 = 2176$$

$$\text{Corrected } \Sigma Y^2: 3010 - 15^2 + 12^2 = 3010 - 225 + 144 = 2929$$

$$\text{Corrected } \Sigma XY: 2460 - (12 \times 15) + (10 \times 12) = 2460 - 180 + 120 = 2400$$

Now, substitute these corrected statistics into the raw data formula:

$$\text{Numerator} = 20(2400) - (198)(237) = 48,000 - 46,926 = 1,074$$

$$\text{Denominator X-term} = 20(2176) - (198)^2 = 43,520 - 39,204 = 4,316$$

$$\text{Denominator Y-term} = 20(2929) - (237)^2 = 58,580 - 56,169 = 2,411$$

$$\text{Corrected } r = 1,074 / \sqrt{[4,316 \times 2,411]} = 1,074 / \sqrt{[10,405,876]} = 1,074 / 3225.814 = 0.3329$$

Interpretation: After correcting the data entry error, the correlation coefficient dropped from an over-inflated 0.45 to a weaker value of $r = 0.3329$. This underscores how sensitive Pearson's coefficient can be to individual data points.

PROBLEM 10: COEFFICIENT OF CONCURRENT DEVIATIONS ANALYSIS

While not as common as Pearson's or Spearman's methods, the **Coefficient of Concurrent Deviations** is a quick, alternative way to look at directionality. It focuses entirely on the signs of the changes from one period to the next. Let's use it to analyze the direction of movement across 6 consecutive periods for market variables X and Y:

X Value: 100, 102, 101, 105, 108, 107

Y Value: 50, 52, 49, 53, 54, 52

SOLUTION

First, we determine the direction of change for each period, marking an increase with a plus sign (+) and a decrease with a minus sign (-). We then look for concurrent deviations, which occur when both variables change in the same direction (both + or both -).

The number of comparative periods is $m = n - 1 = 6 - 1 = 5$.

Period	X	Sign of X Change	Y	Sign of Y Change	Product of Signs
1	100	-	50	-	-
2	102	+	52	+	+ (Concurrent)
3	101	-	49	-	+ (Concurrent)
4	105	+	53	+	+ (Concurrent)
5	108	+	54	+	+ (Concurrent)
6	107	-	52	-	+ (Concurrent)

Counting the concurrent deviations, we find that the signs match in all 5 comparative periods. Therefore, the number of concurrent deviations is $c = 5$.

The formula for the coefficient of concurrent deviations (r_c) is:

$$r_c = \pm \sqrt{[\pm (2c - m) / m]}$$

Substitute our values ($c = 5$, $m = 5$) into the formula:

$$2c - m = 2(5) - 5 = 10 - 5 = 5$$

Since this term is positive, the signs outside and inside the square root remain positive:

$$r_c = + \sqrt{[5 / 5]} = \sqrt{1} = +1.00$$

Interpretation: The coefficient $r_c = +1.00$ indicates that the two market variables move in perfect directional harmony over time.

MODULE 2: REGRESSION ANALYSIS

2.1 Purpose, Logic, and Core Distinction from Correlation

While correlation analysis quantifies the strength and direction of the linear relationship between two variables, it cannot provide a predictive framework. It treats both variables symmetrically and does not establish a mathematical dependency or functional line of best fit. **Regression Analysis** fills this gap. It is a predictive statistical technique used to model and analyze the relationship between a dependent (response) variable and one or more independent (predictor) variables.

The primary goals of regression analysis are estimation and forecasting. By fitting a mathematical equation to the observed data, we can estimate the expected value of the dependent variable based on known values of the independent variable, and quantify the underlying uncertainty of those predictions.

Key Differences Between Correlation and Regression

Diagnostic Criteria	Correlation Analysis	Regression Analysis
Primary Objective	To measure the strength and direction of a linear relationship.	To model a functional relationship for prediction and forecasting.
Variable Roles	Symmetrical. No distinction between independent and dependent variables.	Asymmetrical. Clearly defines independent (predictor) and dependent (response) variables.
Nature of the Metric	A relative, dimensionless index bounded between -1.00 and +1.00.	Absolute equations expressed in the original units of the measured variables.
Cause-and-Effect	Does not imply or establish causal dependency.	Assumes a directional dependency of the response variable on the predictor(s).
Origin/Scale Shifts	Completely unaffected by changes to either origin or scale.	Independent of origin shifts, but highly sensitive to scale changes.

2.2 Formulation of Linear Regression Equations

In simple linear regression, the relationship between two variables is modeled using a straight line. For any bivariate dataset, we can construct two separate regression lines depending on our analytical goal:

1. **The Regression Line of Y on X:** Used when we want to predict or estimate values of the dependent variable **Y** based on known values of the independent variable **X**. The general linear equation is:

$$Y = a + bX$$

Where **a** is the vertical intercept and **b** is the slope of the line (also known as the regression coefficient of Y on X, denoted b_{yx}).

2. **The Regression Line of X on Y:** Used when we want to predict or estimate values of the variable **X** based on known values of **Y**. The general linear equation is:

$$X = c + dY$$

Where **c** is the horizontal intercept and **d** is the slope of the line (also known as the regression coefficient of X on Y, denoted b_{xy}).

2.3 Mathematical Derivation of Least Squares Normal Equations

To find the optimal values for the parameters of our regression lines, we use the method of **Ordinary Least Squares (OLS)**. This technique minimizes the sum of the squared differences (residuals) between the observed values and the values predicted by the regression line.

Derivation for the Line of Y on X

Let Y_i represent an observed value from our dataset, and let $\hat{Y}_i = a + bX_i$ represent the corresponding value predicted by our regression line. The residual error for any given observation is the difference between the two: $e_i = Y_i - \hat{Y}_i = Y_i - (a + bX_i)$.

We want to minimize the total sum of these squared residual errors, which we can define as a function **E**:

$$E(a, b) = \sum e_i^2 = \sum (Y_i - a - bX_i)^2$$

To find the values of **a** and **b** that minimize this function, we take the partial derivatives of **E** with respect to both parameters and set them equal to zero:

$$\partial E / \partial a = -2 \sum (Y_i - a - bX_i) = 0$$

$$\partial E / \partial b = -2 \sum X_i(Y_i - a - bX_i) = 0$$

Simplifying the first partial derivative gives us our first normal equation:

$$\Sigma Y_i - na - b\Sigma X_i = 0 \Rightarrow \Sigma Y = na + b\Sigma X$$

Simplifying the second partial derivative gives us our second normal equation:

$$\Sigma X_i Y_i - a\Sigma X_i - b\Sigma X_i^2 = 0 \Rightarrow \Sigma XY = a\Sigma X + b\Sigma X^2$$

By solving these two **normal equations** simultaneously, we can uniquely determine the intercept **a** and slope **b** for the regression line of Y on X for any given dataset.

Using the exact same optimization process for the regression line of X on Y ($X = c + dY$), we get its corresponding pair of normal equations:

$$\Sigma X = nc + d\Sigma Y$$

$$\Sigma XY = c\Sigma Y + d\Sigma Y^2$$

2.4 Regression Coefficients: Properties and Interpretations

Instead of solving the normal equations from scratch every time, we can express the slopes of our regression lines directly using **regression coefficients**. These coefficients are closely linked to our standard deviations and correlation coefficient.

Deviation Formulations

The regression coefficient of Y on X (b_{yx}) measures the expected change in **Y** for a one-unit change in **X**:

$$b_{yx} = r \times (\sigma_y / \sigma_x) = [n\Sigma XY - (\Sigma X)(\Sigma Y)] / [n\Sigma X^2 - (\Sigma X)^2] = \Sigma xy / \Sigma x^2$$

The regression coefficient of X on Y (b_{xy}) measures the expected change in **X** for a one-unit change in **Y**:

$$b_{xy} = r \times (\sigma_x / \sigma_y) = [n\Sigma XY - (\Sigma X)(\Sigma Y)] / [n\Sigma Y^2 - (\Sigma Y)^2] = \Sigma xy / \Sigma y^2$$

Using these coefficients, we can rewrite our regression equations in a highly useful **mean-deviation form** that passes directly through the mean coordinates ($\bar{X}, \bar{Y}$) of the dataset:

$$\text{Line Y on X: } Y - \bar{Y} = b_{yx}(X - \bar{X})$$

$$\text{Line X on Y: } X - \bar{X} = b_{xy}(Y - \bar{Y})$$

Fundamental Properties of Regression Coefficients

Property 1: Geometric Mean Relationship. The correlation coefficient r is the geometric mean of the two regression coefficients.

$$r = \pm \sqrt{b_{yx} \times b_{xy}}$$

Analytical Proof: Substitute the standard deviation formulas into the product of the two regression coefficients:

$$b_{yx} \times b_{xy} = [r \cdot (\sigma_y / \sigma_x)] \times [r \cdot (\sigma_x / \sigma_y)]$$

Notice that the standard deviation terms cancel out completely:

$$b_{yx} \times b_{xy} = r^2 \times (\sigma_y / \sigma_x) \times (\sigma_x / \sigma_y) = r^2$$

Taking the square root of both sides gives: $r = \pm \sqrt{b_{yx} \times b_{xy}}$. This confirms that the correlation coefficient is the geometric mean of the two regression slopes.

Property 2: Structural Sign Alignment. Both regression coefficients (b_{yx} and b_{xy}) and the correlation coefficient (r) must always share the exact same algebraic sign.

Because standard deviations are always positive numbers, the signs of b_{yx} and b_{xy} are determined entirely by the sign of the correlation coefficient r (or the underlying covariance). If r is negative, both regression coefficients must be negative; if r is positive, both must be positive. It is mathematically impossible for them to have opposing signs.

Property 3: The Magnitude Constraint. If one of the regression coefficients is greater than 1, the other regression coefficient must be less than 1.

Analytical Proof: We know that the squared correlation coefficient cannot exceed 1 ($r^2 \leq 1$). Since $b_{yx} \times b_{xy} = r^2$, it follows that:

$$b_{yx} \times b_{xy} \leq 1$$

If we assume that $b_{yx} > 1$, then for the inequality to hold true, b_{xy} must be less than or equal to $1 / b_{yx}$, which means $b_{xy} < 1$. This property serves as an excellent validation check when calculating regression lines manually.

Property 4: Shift Neutrality. Regression coefficients are unaffected by changes to the origin of the data, but they change systematically when the scale of the data is modified.

If you shift the data by defining new variables $u = (X - a) / h$ and $v = (Y - b) / k$, the new regression coefficients scale as follows:

$$b_{yx} = (k / h) \times b_{vu}$$

$$b_{xy} = (h / k) \times b_{uv}$$

2.5 Standard Error of Estimate and Residual Analysis

The regression line represents the average trend in the data, but individual data points will still vary around this line. To measure the accuracy of our regression model's predictions, we calculate the **Standard Error of Estimate** (denoted $S.E._{yx}$ or S_{yx}). This metric quantifies the reliability of our predictions by measuring the typical spread of the observed data points around the fitted regression line.

The formulas for the Standard Error of Estimate for our two regression lines are:

$$S.E._{yx} = \sqrt{[\Sigma(Y - \hat{Y})^2 / n]} = \sigma_y \times \sqrt{[1 - r^2]}$$

$$S.E._{xy} = \sqrt{[\Sigma(X - \hat{X})^2 / n]} = \sigma_x \times \sqrt{[1 - r^2]}$$

Coefficient of Determination (r^2)

The square of the correlation coefficient, r^2 , is known as the **Coefficient of Determination**. It is a highly useful metric that measures the proportion of the total variation in the dependent variable that can be explained by the independent variable via our regression model.

$$r^2 = \text{Explained Variance} / \text{Total Variance}$$

For example, if $r = 0.90$, then $r^2 = 0.81$. This means that 81% of the variance in the dependent variable is explained by changes in our independent variable. The remaining 19% ($1 - r^2$) is called the **Coefficient of Non-determination**. It represents the unexplained variance—the variation caused by random noise or other external factors not included in our simple linear model.

2.6 Comprehensive Problem Sets on Linear Regression Modeling

PROBLEM 11: NORMAL EQUATIONS FRAMEWORK APPLICATION

A logistics company wants to predict fleet maintenance costs based on total miles driven. The table below shows truck operational hours (X) and monthly maintenance costs (Y, in hundreds of dollars) for 5 vehicles:

Hours (X): 10, 20, 30, 40, 50

Cost (Y): 11, 19, 31, 39, 50

Using the Ordinary Least Squares (OLS) method, find the normal equations, determine the line of best fit for Y on X, and estimate the expected maintenance cost for a vehicle with 35 operational hours.

SOLUTION

Our objective is to find the parameters for the line $Y = a + bX$ using the normal equations:

- 1) $\Sigma Y = na + b\Sigma X$
- 2) $\Sigma XY = a\Sigma X + b\Sigma X^2$

Let's construct our calculation matrix with $n = 5$:

X	Y	X ²	XY
10	11	100	110
20	19	400	380
30	31	900	930
40	39	1600	1560
50	50	2500	2500
$\Sigma=150$	$\Sigma=150$	$\Sigma X^2=5500$	$\Sigma XY=5480$

Substitute these totals into the normal equations:

- 1) $150 = 5a + 150b$
- 2) $5480 = 150a + 5500b$

To isolate and solve for **b**, multiply the first equation by 30 to match the **a** term in the second equation:

$$30 \times (150 = 5a + 150b) \Rightarrow 4500 = 150a + 4500b$$

Subtract this new equation from the second normal equation:

$$(5480 = 150a + 5500b) - (4500 = 150a + 4500b)$$

$$980 = 1000b \Rightarrow b = 980 / 1000 = 0.98$$

Now, substitute $b = 0.98$ back into the first equation to solve for a :

$$150 = 5a + 150(0.98) \Rightarrow 150 = 5a + 147$$

$$5a = 150 - 147 = 3 \Rightarrow a = 3 / 5 = 0.60$$

This gives us our finalized linear regression equation for Y on X:

$$\hat{Y} = 0.60 + 0.98X$$

To estimate the maintenance cost for $X = 35$ hours, substitute the value into our equation:

$$\hat{Y} = 0.60 + 0.98(35) = 0.60 + 34.30 = 34.90$$

Interpretation: The base monthly maintenance cost is \$60 (intercept $a = 0.60$). For every additional operational hour, costs are expected to increase by \$98 (slope $b = 0.98$). For a truck with 35 operational hours, the predicted maintenance cost is \$3,490 ($34.90 \times \100).

PROBLEM 12: DUAL REGRESSION LINE CALCULATIONS VIA DEVIATION FORM

A corporate study evaluates the relationship between an employee's years of experience (X) and their monthly performance score (Y). Given the following paired data for 5 employees, calculate both regression lines (Y on X and X on Y) using the mean deviation method.

Experience (X): 1, 2, 3, 4, 5

Performance (Y): 10, 15, 18, 22, 30

SOLUTION

First, find the arithmetic means for both variables with $n = 5$:

$$\Sigma X = 15 \Rightarrow \bar{X} = 15 / 5 = 3$$

$$\Sigma Y = 95 \Rightarrow \bar{Y} = 95 / 5 = 19$$

Next, construct our deviation table using $x = X - 3$ and $y = Y - 19$:

X	Y	$x = X - 3$	$y = Y - 19$	x^2	y^2	xy
1	10	-2	-9	4	81	18
2	15	-1	-4	1	16	4
3	18	0	-1	0	1	0
4	22	1	3	1	9	3
5	30	2	11	4	121	22
$\Sigma=15$	$\Sigma=95$	$\Sigma x=0$	$\Sigma y=0$	$\Sigma x^2=10$	$\Sigma y^2=228$	$\Sigma xy=47$

Now, calculate the regression coefficients b_{yx} and b_{xy} :

$$b_{yx} = \Sigma xy / \Sigma x^2 = 47 / 10 = 4.7$$

$$b_{xy} = \Sigma xy / \Sigma y^2 = 47 / 228 = 0.2061$$

Using these coefficients, we establish our two regression lines:

1. Regression Equation of Y on X:

$$Y - \bar{Y} = b_{yx}(X - \bar{X}) \Rightarrow Y - 19 = 4.7(X - 3)$$

$$Y = 19 + 4.7X - 14.1 \Rightarrow Y = 4.9 + 4.7X$$

2. Regression Equation of X on Y:

$$X - \bar{X} = b_{xy}(Y - \bar{Y}) \Rightarrow X - 3 = 0.2061(Y - 19)$$

$$X = 3 + 0.2061Y - 3.9159 \Rightarrow X = -0.9159 + 0.2061Y$$

Validation Check: Let's verify the magnitude constraint. Since $b_{yx} = 4.7$ (which is greater than 1), $b_{xy} = 0.2061$ must be less than 1, which it is. Furthermore, we can find the implied correlation coefficient: $r = \sqrt{[4.7 \times 0.2061]} = \sqrt{0.96867} = +0.9842$. This reflects a very strong positive relationship.

PROBLEM 13: DERIVING SUMMARY VALUES FROM INTERSECTING LINES

A researcher modeling marketing metrics extracts two linear regression equations:

Line 1: $8X - 10Y + 66 = 0$

Line 2: $40X - 18Y - 214 = 0$

Identify which equation represents the line of Y on X and which represents X on Y, find the mean values ($\bar{X}$ and $\bar{Y}$), and calculate the correlation coefficient r .

SOLUTION

First, find the mean values ($\bar{X}$, $\bar{Y}$). Because both regression lines must intersect exactly at the coordinates of their arithmetic means, we can find $\bar{X}$ and $\bar{Y}$ by solving the two equations simultaneously:

$$1) 8X - 10Y = -66$$

$$2) 40X - 18Y = 214$$

Multiply the first equation by 5 to align the X terms:

$$5 \times (8X - 10Y = -66) \Rightarrow 40X - 50Y = -330$$

Subtract this new equation from the second equation:

$$(40X - 18Y = 214) - (40X - 50Y = -330)$$

$$32Y = 544 \Rightarrow \bar{Y} = 544 / 32 = 17$$

Substitute $\bar{Y} = 17$ back into the first equation to solve for $\bar{X}$:

$$8X - 10(17) = -66 \Rightarrow 8X - 170 = -66$$

$$8X = 104 \Rightarrow \bar{X} = 104 / 8 = 13$$

Thus, our arithmetic means are $\bar{X} = 13$ and $\bar{Y} = 17$.

Next, we need to correctly identify which line is which. To do this, we will make a guess, calculate the resulting slopes, and check if they satisfy our magnitude constraint ($b_{yx} \times b_{xy} \leq 1$).

Let's assume Line 1 is the regression line of Y on X:

$$8X - 10Y + 66 = 0 \Rightarrow 10Y = 8X + 66 \Rightarrow Y = 0.8X + 6.6$$

This gives a slope of $b_{yx} = 0.8$.

Following this assumption, Line 2 must be the regression line of X on Y:

$$40X - 18Y - 214 = 0 \Rightarrow 40X = 18Y + 214 \Rightarrow X = (18/40)Y + (214/40) \Rightarrow X = 0.45Y + 5.35$$

This gives a slope of $b_{xy} = 0.45$.

Now, validate our assumption using the magnitude constraint:

$$b_{yx} \times b_{xy} = 0.8 \times 0.45 = 0.36$$

Since the product $0.36 \leq 1$, our assumption is correct! If the product had been greater than 1, we would have reversed our assumption.

Finally, solve for the correlation coefficient r :

$$r = \sqrt{[b_{yx} \times b_{xy}]} = \sqrt{0.36} = 0.60$$

Since both regression slopes are positive, r must also be positive: $r = +0.60$.

PROBLEM 14: STANDARD ERROR OF ESTIMATE CALCULATION MATRIX

Using the data and results from Problem 11 ($\hat{Y} = 0.60 + 0.98X$), calculate the Standard Error of Estimate ($S.E._{yx}$) by finding the individual residuals for each observation.

SOLUTION

Let's build a matrix to calculate the predicted values ($\hat{Y}$) and residual errors ($Y - \hat{Y}$) for each data point:

X	Y	Predicted $\hat{Y} = 0.60 + 0.98X$	Residual ($Y - \hat{Y}$)	Squared Residual ($Y - \hat{Y})^2$
10	11	$0.60 + 0.98(10) = 10.40$	$11 - 10.40 = 0.60$	0.3600
20	19	$0.60 + 0.98(20) = 20.20$	$19 - 20.20 = -1.20$	1.4400
30	31	$0.60 + 0.98(30) = 30.00$	$31 - 30.00 = 1.00$	1.0000
40	39	$0.60 + 0.98(40) = 39.80$	$39 - 39.80 = -0.80$	0.6400
50	50	$0.60 + 0.98(50) = 50.60$	$50 - 50.60 = -0.60$	0.3600
-	-	-	$\Sigma(Y - \hat{Y}) = 0.00$	$\Sigma(Y - \hat{Y})^2 = 3.8000$

Substitute the sum of the squared residuals into the Standard Error formula with $n = 5$:

$$S.E._{yx} = \sqrt{[\Sigma(Y - \hat{Y})^2 / n]} = \sqrt{[3.80 / 5]} = \sqrt{0.76} = 0.8718$$

Interpretation: The Standard Error of Estimate is 0.8718 (or \$87.18). This small error value confirms that our linear regression model fits the observed data very well, providing highly accurate predictions.

PROBLEM 15: RECONSTRUCTING STRUCTURAL VARIANCES FROM SUMMARY INPUTS

An executive HR dashboard provides the following summary metrics for a corporate department:

$\bar{X} = 40$, $\bar{Y} = 50$, $b_{yx} = 1.20$, and $b_{xy} = 0.30$.

Calculate the correlation coefficient r , and find the variance ratio between the two variables (σ_y^2 / σ_x^2).

SOLUTION

First, find the correlation coefficient r by calculating the geometric mean of the two regression slopes:

$$r = \sqrt{[b_{yx} \times b_{xy}]} = \sqrt{[1.20 \times 0.30]} = \sqrt{0.36} = 0.60$$

Since both slopes are positive, r is positive: $r = +0.60$.

Next, we use our regression coefficient formulas to isolate the variances:

$$b_{yx} = r \times (\sigma_y / \sigma_x) \Rightarrow 1.20 = 0.60 \times (\sigma_y / \sigma_x)$$

Dividing both sides by 0.60 gives the standard deviation ratio:

$$\sigma_y / \sigma_x = 1.20 / 0.60 = 2.0$$

To find the variance ratio, we simply square this result:

$$\sigma_y^2 / \sigma_x^2 = (2.0)^2 = 4.0$$

Interpretation: The variation in variable Y is four times greater than the variation in variable X .

PROBLEM 16: APPLIED ESTIMATION VIA STEP-DEVIATION SCALE CONVERSIONS

A manufacturing group monitors production yield (Y) based on chemical catalyst weight (X). Because the raw values are large, the analyst applies step-deviations using assumed means ($A_x = 150$, $A_y = 400$) and scaling factors ($h = 10$, $k = 25$), defining the new variables as $u = (X - 150)/10$ and $v = (Y - 400)/25$.

The scaled data yields the following summary statistics: $n = 10$, $\Sigma u = 20$, $\Sigma v = 30$, $\Sigma u^2 = 180$, and $\Sigma uv = 140$.

Determine the true regression coefficient b_{yx} and construct the final regression equation for Y on X .

SOLUTION

First, find the regression coefficient for the scaled variables, b_{vu} :

$$b_{vu} = [n\Sigma uv - (\Sigma u)(\Sigma v)] / [n\Sigma u^2 - (\Sigma u)^2] = [10(140) - (20)(30)] / [10(180) - (20)^2]$$

$$b_{vu} = [1400 - 600] / [1800 - 400] = 800 / 1400 = 0.5714$$

Next, convert this scaled coefficient back to our original units using our scaling factors ($h = 10$, $k = 25$):

$$b_{yx} = (k / h) \times b_{vu} = (25 / 10) \times 0.5714 = 2.5 \times 0.5714 = 1.4285$$

Now, find the true arithmetic means ($\bar{X}$ and $\bar{Y}$) by reversing the original transformation equations:

$$\bar{u} = \Sigma u / n = 20 / 10 = 2 \Rightarrow \bar{X} = A_x + (\bar{u} \times h) = 150 + (2 \times 10) = 170$$

$$\bar{v} = \Sigma v / n = 30 / 10 = 3 \Rightarrow \bar{Y} = A_y + (\bar{v} \times k) = 400 + (3 \times 25) = 475$$

Finally, substitute the true means and coefficient into the mean-deviation equation:

$$Y - \bar{Y} = b_{yx}(X - \bar{X}) \Rightarrow Y - 475 = 1.4285(X - 170)$$

$$Y = 475 + 1.4285X - 242.845 \Rightarrow Y = 232.155 + 1.4285X$$

Interpretation: The final model is $Y = 232.155 + 1.4285X$, allowing us to accurately predict production yield based on catalyst weight in the original scale metrics.

PROBLEM 17: ANGLE DERIVATION BETWEEN INTERSECTING LINES

Show mathematically how the angle between our two linear regression lines is related to the correlation coefficient r . Given $\sigma_x = 3$, $\sigma_y = 4$, and $r = 0.60$, calculate the precise angle of intersection.

SOLUTION

The slopes of our two regression lines are given by $m_1 = b_{yx} = r(\sigma_y / \sigma_x)$ and $m_2 = 1 / b_{xy} = \sigma_y / (r \sigma_x)$.

The trigonometric identity for the angle θ between two intersecting lines is:

$$\tan(\theta) = | (m_2 - m_1) / (1 + m_1 m_2) |$$

Substituting our slope formulations into this identity gives the standard relationship:

$$\tan(\theta) = [(1 - r^2) / r] \times [(\sigma_x \sigma_y) / (\sigma_x^2 + \sigma_y^2)]$$

Now, let's substitute our known numerical values into this equation:

$$1 - r^2 = 1 - 0.60^2 = 1 - 0.36 = 0.64$$

$$\sigma_x \sigma_y = 3 \times 4 = 12$$

$$\sigma_x^2 + \sigma_y^2 = 3^2 + 4^2 = 9 + 16 = 25$$

Solve for $\tan(\theta)$:

$$\tan(\theta) = [0.64 / 0.60] \times [12 / 25] = 1.0667 \times 0.48 = 0.5120$$

$$\theta = \arctan(0.5120) = 27.11^\circ$$

Analytical Insights: If $r = \pm 1$, then $\tan(\theta) = 0$, meaning $\theta = 0^\circ$. The two regression lines overlap completely because a perfect correlation means both lines are identical. If $r = 0$, the term becomes

undefined, meaning $\theta = 90^\circ$. The two lines cross at a perfect right angle because the variables share zero linear relationship.

PROBLEM 18: UNEXPLAINED VARIANCE DECOMPOSITION ANALYSIS

A regression analysis modeling corporate energy consumption yields a correlation coefficient of $r = 0.80$. The total measured variance of the energy consumption variable is $\sigma_y^2 = 150$. Deconstruct this total variance into its explained and unexplained components.

SOLUTION

We use the Coefficient of Determination (r^2) to split the variance into its components:

$$r^2 = 0.80^2 = 0.64 \text{ (64\% of the variance is explained)}$$

$$1 - r^2 = 1 - 0.64 = 0.36 \text{ (36\% of the variance is unexplained)}$$

Now, calculate the absolute variance values:

$$\text{Explained Variance} = r^2 \times \sigma_y^2 = 0.64 \times 150 = 96$$

$$\text{Unexplained Variance} = (1 - r^2) \times \sigma_y^2 = 0.36 \times 150 = 54$$

Validation Check: The sum of the components ($96 + 54 = 150$) equals our total variance. This confirms that 96 units of variance are explained by our regression model, while 54 units remain unexplained noise or residual error.

PROBLEM 19: EVALUATION OF NON-LINEARITY TRIGGERS VIA RESIDUAL SPREAD

An analyst models a business process using linear regression, but suspects the true relationship might be non-linear. They calculate the following residuals ($e = Y - \hat{Y}$) across 6 sequential levels of the independent variable X:

X Value: 1, 2, 3, 4, 5, 6

Residual (e): -5, -2, 4, 5, 1, -3

Evaluate this residual pattern to determine if a linear model is appropriate.

SOLUTION

A key assumption of linear regression is that residuals should be randomly distributed around zero. If we look at the sequence of our residuals, we notice a distinct, non-random pattern:

- At low values of X (1 and 2), the residuals are negative.

- At middle values of X (3, 4, and 5), the residuals become positive.
- At the highest value of X (6), the residual drops back to negative.

Instead of random noise, the residuals follow an inverted U-shaped curve.

Diagnostic Conclusion: This systematic, curved pattern in the residuals indicates that the true relationship between the variables is non-linear (curvilinear). A simple linear model is not appropriate here; the analyst should use a quadratic or polynomial regression model ($Y = a + bX + cX^2$) to properly capture the relationship.

PROBLEM 20: MATHEMATICAL PROOF: THE MEAN COORDINATES INTERSECTION RULE

Prove analytically that the regression line of Y on X ($Y = a + bX$) must always pass exactly through the mean coordinates ($\bar{X}$, $\bar{Y}$) of the dataset.

SOLUTION

Let's revisit our first OLS normal equation derived in Section 2.3:

$$\Sigma Y = na + b\Sigma X$$

To express this equation in terms of means, we divide both sides by the total sample size n :

$$\Sigma Y / n = (na / n) + b(\Sigma X / n)$$

By definition, the mean of Y is $\bar{Y} = \Sigma Y / n$ and the mean of X is $\bar{X} = \Sigma X / n$. Substituting these terms into our equation gives:

$$\bar{Y} = a + b\bar{X}$$

This completed equation proves that if you substitute the mean value $\bar{X}$ into the regression line, the resulting predicted value will always equal the true mean value $\bar{Y}$. Therefore, the line of best fit must pass directly through the point ($\bar{X}$, $\bar{Y}$) for any linear dataset.

MODULE 3: BUSINESS FORECASTING AND TIME SERIES ANALYSIS

3.1 Executive Framework of Business Forecasting

In modern operational management, **Business Forecasting** is the structured process of projecting future business performance, market dynamics, and economic conditions using historical and current data. It provides the essential groundwork for strategic planning, corporate budgeting, supply chain optimization, and risk management.

Every forecasting process must balance historical patterns with future uncertainties. While automated forecasting systems can quickly process vast amounts of data, they operate under a fundamental assumption: that the structural patterns observed in the past will continue into the future. Therefore, effective business forecasting requires combining systematic quantitative algorithms with qualitative human insights to adjust for sudden market shifts, regulatory changes, or disruptive innovations.

3.2 Taxonomy of Forecasting Methods

Business forecasting techniques are broadly divided into two primary categories: qualitative frameworks and quantitative models.

1. Qualitative (Judgmental) Methodology

Qualitative methods rely on human expertise, intuition, and experience rather than historical numerical data. They are typically used when historical data is unavailable, scarce, or no longer relevant (such as launching an entirely new product line or navigating a major structural market shift).

- **Delphi Method:** A structured, iterative forecasting technique that relies on a panel of independent experts. An internal facilitator sends out a series of questionnaires to the experts, who provide their forecasts anonymously. After each round, the facilitator provides an anonymous summary of the experts' forecasts and reasons. The experts are then encouraged to revise their earlier answers in light of the replies of other members of the panel. This process continues through several rounds until the group converges toward a consensus forecast.
- **Jury of Executive Opinion:** A method that combines the high-level views, insights, and estimates of top executives from various departments (such as Finance, Sales, and Production) to form a unified corporate forecast.

- **Sales Force Composite:** A bottom-up approach where each sales representative estimates future demand within their specific territory. These individual projections are then aggregated and adjusted at the regional and national levels to build an overall sales forecast.

2. Quantitative Methodology

Quantitative methods use mathematical models to analyze historical numerical data. They operate under the assumption that historical patterns contain valuable signals that can be projected forward to predict future values. These methods are further divided into two types: **causal models** (such as regression analysis, which models relationships between different variables) and **time series models** (which focus entirely on patterns within a single variable recorded sequentially over time).

3.3 Introduction to Time Series Analysis

A **Time Series** is a sequence of numerical data points recorded at uniform, successive intervals over time (e.g., daily stock prices, monthly sales revenues, or annual crop yields). Formally, it is modeled as a functional dependency: $Y = f(t)$, where Y is the measured value and t represents time.

The primary goal of time series analysis is to break down the historical data into its underlying components, isolate systematic patterns from random noise, and project those patterns forward to create accurate forecasts.

Mathematical Structural Models

To analyze a time series, we must choose how to combine its underlying components. There are two primary mathematical models used for this decomposition:

1. **The Additive Model:** Assumes that the observed value is simply the sum of its independent components. This model is appropriate when the seasonal and cyclical variations remain relatively constant in magnitude, regardless of the overall trend level.

$$Y_t = T_t + S_t + C_t + I_t$$

2. **The Multiplicative Model:** Assumes that the components change proportionally based on the overall trend. In this model, the seasonal and cyclical variations grow or shrink in size as the trend scales upward or downward. This configuration is much more common in economic and business applications.

$$Y_t = T_t \times S_t \times C_t \times I_t$$

Where T represents the trend value, and S , C , I are expressed as percentage indexes or multipliers.

3.4 Component Breakdown of a Time Series

A time series dataset is driven by four distinct statistical components:

Component	Operational Definition	Typical Duration	Real-World Example
Secular Trend (T)	The long-term, smooth directional movement of the data over an extended period.	Years or decades.	Increases in global smartphone adoption; long-term population growth.
Seasonal Variation (S)	Systematic, rhythmic fluctuations that repeat regularly within a single year.	Within 1 year (daily, weekly, monthly, quarterly).	Spikes in retail sales during the holidays; surges in electricity demand during summer.
Cyclical Variation (C)	Long-term, wave-like fluctuations driven by broader economic and business cycles.	Typically 2 to 10 years.	Periods of economic expansion, peak, recession, and recovery.
Irregular Variation (I)	Unpredictable, random fluctuations caused by sudden, unforeseen events.	Short-term, erratic.	Supply chain disruptions caused by natural disasters, strikes, or sudden political events.

3.5 Quantitative Methodology: Moving Averages Method

The **Moving Averages Method** is a smoothing technique used to remove short-term seasonal and irregular fluctuations from a time series, making it easier to see the underlying secular trend. It works by replacing each data point with the arithmetic mean of a specified number of adjacent periods.

The choice of the moving average period is critical:

- An **odd-period moving average** (e.g., 3-year or 5-year) is straightforward because the calculated mean aligns perfectly with the exact center of the time interval.
- An **even-period moving average** (e.g., 4-quarter or 12-month) requires a two-step process called **centering**. Because the mean of an even number of periods falls between two time points, we must calculate a second moving average of the results to align the trend values precisely with the original time periods.

3.6 Quantitative Methodology: OLS Least Squares Trend Fitting

The most precise way to model the long-term trend in a time series is by fitting a mathematical equation to the data using the **Method of Least Squares**. This is identical to the OLS regression process discussed in Module 2, but here the independent variable is time (t).

The Linear Trend Equation

A linear trend is modeled using the equation:

$$Y_t = a + bt$$

Where Y_t is the trend value at time t , a is the intercept, and b is the annual growth or decline rate. The normal equations to solve for these parameters are:

$$\Sigma Y = na + b\Sigma t$$

$$\Sigma tY = a\Sigma t + b\Sigma t^2$$

The Time-Centering Transformation Shortcut

We can simplify these calculations significantly by shifting the origin of our time variable so that the sum of the time periods equals zero ($\Sigma t = 0$). This is called **time-centering**:

- For an odd number of years, set $t = 0$ at the middle year, using increments of ± 1 for adjacent years.
- For an even number of years, set the origin between the two middle years, using increments of ± 1 for each half-year (which simplifies to odd integers: ..., -5, -3, -1, +1, +3, +5, ...).

When $\Sigma t = 0$, the normal equations simplify beautifully, allowing us to calculate a and b directly without solving simultaneous equations:

$$a = \Sigma Y / n$$

$$b = \Sigma tY / \Sigma t^2$$

3.7 Comprehensive Problem Sets on Time Series Analysis

PROBLEM 21: ODD-PERIOD MOVING AVERAGES SMOOTHING MODEL

A retail brand records its annual sales revenue (Y, in millions of dollars) over a 7-year operational span. Use a 3-year moving average model to smooth the data and isolate the underlying trend.

Year: 2020, 2021, 2022, 2023, 2024, 2025, 2026

Revenue (Y): 12, 15, 14, 18, 17, 22, 21

SOLUTION

Since we are using a 3-year period (an odd number), each moving average will align perfectly with the middle year of its 3-year window. Let's calculate the 3-year moving totals and then divide by 3 to find the moving averages:

Year	Revenue (Y)	3-Year Moving Total	3-Year Moving Average (Trend)
2020	12	-	- (No preceding data)
2021	15	$12 + 15 + 14 = 41$	$41 / 3 = 13.67$
2022	14	$15 + 14 + 18 = 47$	$47 / 3 = 15.67$
2023	18	$14 + 18 + 17 = 49$	$49 / 3 = 16.33$
2024	17	$18 + 17 + 22 = 57$	$57 / 3 = 19.00$
2025	22	$17 + 22 + 21 = 60$	$60 / 3 = 20.00$
2026	21	-	- (No succeeding data)

Interpretation: The moving average successfully smooths out the short-term fluctuations (such as the dip in 2022), revealing a clear, steady upward trend in sales revenue from \$13.67 million to \$20.00 million.

PROBLEM 22: EVEN-PERIOD MOVING AVERAGES WITH CENTERING MATRIX

An e-commerce business tracks its quarterly sales volumes (Y, in thousands of orders) over two years. Because quarterly data has an even number of periods per year, use a 4-quarter moving average with

centering to isolate the trend.

2024: Q1=10, Q2=14, Q3=12, Q4=18 | **2025:** Q1=13, Q2=19, Q3=16, Q4=24

SOLUTION

Because a 4-quarter moving total falls between periods, we must calculate the 4-quarter moving totals, then calculate the 2-period moving totals of those results, and finally divide by 8 to get our centered moving averages:

Year & Qtr	Sales (Y)	4-Quarter Moving Total	2-Period Centered Total	Centered 4-Qtr Moving Avg
2024 Q1	10	-	-	-
2024 Q2	14	-	-	-
2024 Q3	12	$10+14+12+18 = 54$	-	-
2024 Q4	18	$14+12+18+13 = 57$	$54 + 57 = 111$	$111 / 8 = 13.875$
2025 Q1	13	$12+18+13+19 = 62$	$57 + 62 = 119$	$119 / 8 = 14.875$
2025 Q2	19	$18+13+19+16 = 66$	$62 + 66 = 128$	$128 / 8 = 16.000$
2025 Q3	16	$13+19+16+24 = 72$	$66 + 72 = 138$	$138 / 8 = 17.250$
2025 Q4	24	-	-	-

Interpretation: The centered moving average successfully removes quarterly seasonal patterns, revealing a steady underlying growth rate from 13.875 to 17.250 thousand orders.

PROBLEM 23: LINEAR TREND FITTING VIA TIME-CENTERING SHORTCUT (ODD YEARS)

A manufacturing plant records its annual production output (Y, in thousands of tons) over 5 consecutive years. Fit a linear trend line using the time-centering shortcut method, and forecast the expected output for the year 2028.

Year: 2022, 2023, 2024, 2025, 2026

Output (Y): 45, 52, 50, 58, 65

SOLUTION

Since we have 5 years (an odd number), we can center our time variable by setting the middle year (2024) as our origin ($t = 0$). This ensures that $\Sigma t = 0$, allowing us to use our simplified shortcut formulas with $n = 5$.

Year	Output (Y)	Centered Time (t)	t^2	tY
2022	45	-2	4	-90
2023	52	-1	1	-52
2024	50	0	0	0
2025	58	1	1	58
2026	65	2	4	130
Total	$\Sigma Y = 270$	$\Sigma t = 0$	$\Sigma t^2 = 10$	$\Sigma tY = 46$

Calculate the intercept a and slope b using the shortcut formulas:

$$a = \Sigma Y / n = 270 / 5 = 54$$

$$b = \Sigma tY / \Sigma t^2 = 46 / 10 = 4.6$$

This gives us our finalized trend line equation:

$$Y_t = 54 + 4.6t$$

To forecast for the year 2028, determine its corresponding value of t . Since 2026 corresponds to $t = 2$, the year 2028 corresponds to $t = 4$. Substitute $t = 4$ into our trend equation:

$$Y_{2028} = 54 + 4.6(4) = 54 + 18.4 = 72.4$$

Interpretation: The base production level at our 2024 origin is 54,000 tons, with a steady annual growth rate of 4,600 tons (slope $b = 4.6$). The forecasted production output for the year 2028 is 72,400 tons.

PROBLEM 24: LINEAR TREND FITTING FOR AN EVEN NUMBER OF YEARS

Fit a linear trend line to the following dataset showing annual customer acquisitions (Y) over 6 consecutive years, and forecast the expected acquisitions for the year 2027.

Year: 2021, 2022, 2023, 2024, 2025, 2026

Acquisitions (Y): 200, 215, 220, 240, 255, 280

SOLUTION

Since we have 6 years (an even number), our time origin falls directly between the two middle years (2023 and 2024). To avoid fractional numbers while keeping $\Sigma t = 0$, we set our centered time units in half-year steps using odd integers: ..., -5, -3, -1, +1, +3, +5, ... with $n = 6$.

Year	Acquisitions (Y)	Centered Time (t)	t^2	tY
2021	200	-5	25	-1000
2022	215	-3	9	-645
2023	220	-1	1	-220
2024	240	1	1	240
2025	255	3	9	765
2026	280	5	25	1400
Total	$\Sigma Y = 1410$	$\Sigma t = 0$	$\Sigma t^2 = 70$	$\Sigma tY = 540$

Calculate the trend line parameters:

$$a = \Sigma Y / n = 1410 / 6 = 235$$

$$b = \Sigma tY / \Sigma t^2 = 540 / 70 = 7.7143$$

This gives us our trend line equation based on our half-year time units:

$$Y_t = 235 + 7.7143t$$

To forecast for the year 2027, determine its corresponding value of t . Following our sequence where 2026 corresponds to $t = 5$ (representing 2.5 years past the origin), the year 2027 represents 3.5 years past the origin, which corresponds to $t = 7$. Substitute $t = 7$ into our equation:

$$Y_{2027} = 235 + 7.7143(7) = 235 + 54.0 = 289.0$$

Interpretation: The trend line models steady growth, projecting that customer acquisitions will reach 289.0 units in 2027.

PROBLEM 25: NON-LINEAR TREND ANALYSIS: FITTING A QUADRATIC CURVE

A technology startup experiences rapid, accelerating growth in monthly active users (Y, in thousands). Fit a quadratic (parabolic) trend curve ($Y = a + bt + ct^2$) to the following 5 periods, and estimate users for period 6.

Period: 1, 2, 3, 4, 5

Users (Y): 10, 15, 25, 40, 62

SOLUTION

Let's center our time variable by setting the middle period (3) as our origin ($t = 0$) to simplify the equations. This ensures that $\Sigma t = 0$ and $\Sigma t^3 = 0$.

The normal equations for a quadratic curve when $\Sigma t = 0$ are:

1) $\Sigma Y = na + c\Sigma t^2$

2) $\Sigma tY = b\Sigma t^2 \Rightarrow b = \Sigma tY / \Sigma t^2$

3) $\Sigma t^2Y = a\Sigma t^2 + c\Sigma t^4$

Period	Users (Y)	Centered t	t ²	t ⁴	tY	t ² Y
1	10	-2	4	16	-20	40
2	15	-1	1	1	-15	15
3	25	0	0	0	0	0
4	40	1	1	1	40	40
5	62	2	4	16	124	248
Σ	ΣY=152	Σt=0	Σt²=10	Σt⁴=34	ΣtY=129	Σt²Y=343

First, solve for the linear slope coefficient **b** directly from equation 2:

$$b = 129 / 10 = 12.9$$

Next, substitute our remaining totals into equations 1 and 3 to solve for **a** and **c**:

1) $152 = 5a + 10c$

3) $343 = 10a + 34c$

Multiply equation 1 by 2 to align the **a** terms:

$$2 \times (152 = 5a + 10c) \Rightarrow 304 = 10a + 20c$$

Subtract this from equation 3:

$$(343 = 10a + 34c) - (304 = 10a + 20c)$$

$$39 = 14c \Rightarrow c = 39 / 14 = 2.7857$$

Now, substitute $c = 2.7857$ back into equation 1 to solve for a :

$$152 = 5a + 10(2.7857) \Rightarrow 152 = 5a + 27.857$$

$$5a = 124.143 \Rightarrow a = 124.143 / 5 = 24.8286$$

This gives us our finalized quadratic trend curve equation:

$$\hat{Y}_t = 24.8286 + 12.9t + 2.7857t^2$$

To estimate users for period 6, find its corresponding value of t . Since period 5 corresponds to $t = 2$, period 6 corresponds to $t = 3$. Substitute $t = 3$ into our quadratic equation:

$$\hat{Y}_{\text{period 6}} = 24.8286 + 12.9(3) + 2.7857(3)^2 = 24.8286 + 38.70 + 25.0713 = 88.60$$

Interpretation: The positive value for our quadratic term ($c = 2.7857$) confirms that user growth is accelerating over time. The estimated number of active users for period 6 is 88,600.

PROBLEM 26: SEASONAL INDEX CALCULATION VIA RATIO-TO-MOVING-AVERAGE METHOD

An enterprise wants to calculate seasonal indexes for its quarterly sales data using the Ratio-to-Moving-Average method. The table below shows the observed quarterly sales alongside the centered 4-quarter moving averages calculated in Problem 22:

2024 Q4: Obs Sales = 18, Centered MA = 13.875

2025 Q1: Obs Sales = 13, Centered MA = 14.875

2025 Q2: Obs Sales = 19, Centered MA = 16.000

2025 Q3: Obs Sales = 16, Centered MA = 17.250

Calculate the quarterly seasonal ratios and determine the final seasonal indexes.

SOLUTION

First, we calculate the seasonal ratio for each quarter by dividing the observed sales by the corresponding centered moving average, and multiplying by 100: **Ratio = (Observed / Moving Average) × 100**.

- **2024 Q4 Ratio:** $(18 / 13.875) \times 100 = 129.73\%$
- **2025 Q1 Ratio:** $(13 / 14.875) \times 100 = 87.39\%$
- **2025 Q2 Ratio:** $(19 / 16.000) \times 100 = 118.75\%$
- **2025 Q3 Ratio:** $(16 / 17.250) \times 100 = 92.75\%$

In a comprehensive dataset spanning multiple years, we would compute the average ratio for each quarter across all years. For this problem, let's assume these calculated values represent our base quarterly averages:
Q1 = 87.39% | Q2 = 118.75% | Q3 = 92.75% | Q4 = 129.73%

Next, check if the sum of our quarterly averages equals exactly 400% (since there are 4 quarters in a year):

$$\text{Sum} = 87.39 + 118.75 + 92.75 + 129.73 = 428.62\%$$

Since the sum is not exactly 400%, we must apply a correction factor to normalize our values: **Correction Factor** = $400 / 428.62 = 0.93322$. Multiply each quarterly average by this factor to get our final seasonal indexes:

- **Final Q1 Index:** $87.39 \times 0.93322 = 81.56$
- **Final Q2 Index:** $118.75 \times 0.93322 = 110.82$
- **Final Q3 Index:** $92.75 \times 0.93322 = 86.56$
- **Final Q4 Index:** $129.73 \times 0.93322 = 121.06$

Interpretation: These seasonal indexes reveal clear, predictable fluctuations within each year: sales perform weakest in the first quarter (18.44% below the baseline trend) and peak sharply in the fourth quarter (21.06% above the baseline trend).

PROBLEM 27: SHIFTING THE TREND ORIGIN TO A NEW TARGET YEAR

A corporate finance team uses a linear trend equation to model annual operational expenditures: $Y_t = 145 + 12.5t$, where the origin is set at the year 2020 ($t = 0$). Shift the origin of this trend line to the year 2025.

SOLUTION

To shift the origin of a trend line forward, we look at the distance between our old origin and our new target year. The year 2025 is exactly 5 years ahead of our old origin (2020).

Therefore, we can define our new time variable as $t' = t + 5$, which means the old time variable can be rewritten as: $t = t' - 5$.

Substitute this expression into our original trend line equation:

$$Y = 145 + 12.5(t' - 5)$$

$$Y = 145 + 12.5t' - 62.5 = 82.5 + 12.5t'$$

Interpretation: The updated trend equation with its origin set at 2025 is $Y_t = 82.5 + 12.5t'$. Notice that the annual growth rate (slope $b = 12.5$) remains unchanged; only the baseline intercept is updated to reflect the higher cost levels in 2025.

PROBLEM 28: CONVERTING AN ANNUAL LINEAR TREND TO MONTHLY INCREMENTS

An industrial conglomerate models its annual corporate production volume using the linear trend line: $Y = 1200 + 144t$, where the origin is set at the year 2022. Convert this annual trend equation into a monthly trend equation.

SOLUTION

To convert an annual trend line to a monthly trend line, we must adjust both the intercept and the slope to match the monthly scale:

1. Adjusting the Intercept (a): Since the annual intercept represents the total volume for an entire year, we divide it by 12 to find the average monthly baseline volume:

$$\text{Monthly Intercept} = 1200 / 12 = 100$$

2. Adjusting the Slope (b): The annual slope represents the growth rate from one year to the next. To find how much the trend grows from one month to the next, we divide by $12 \times 12 = 144$ (once to convert the annual increment to a monthly increment, and a second time to account for the change in our time units from years to months):

$$\text{Monthly Slope} = 144 / 144 = 1.0$$

This gives us our finalized monthly trend equation:

$$Y_{\text{monthly}} = 100 + 1.0t$$

Interpretation: In our monthly model, the baseline volume starts at 100 units with a steady monthly growth trend of 1.0 unit.

PROBLEM 29: EXPONENTIAL TREND FITTING VIA LOGARITHMIC CONVERSIONS

A retail brand expands rapidly, and its annual revenues (Y , in millions) follow an exponential curve: $Y = a \cdot b^t$. Using logarithms, fit an exponential trend curve to the following 3 periods:

Period (t): -1, 0, 1

Revenue (Y): 4, 12, 36

SOLUTION

First, take the logarithm of both sides of our exponential equation to transform it into a linear form:

$$\log(Y) = \log(a) + t \cdot \log(b)$$

Let's define our transformed variables as $Y' = \log(Y)$, $a' = \log(a)$, and $b' = \log(b)$. This gives us a standard linear equation: $Y' = a' + b't$. Since our time variable is already centered ($\Sigma t = 0$), we can solve for a' and b' directly using our shortcut formulas with $n = 3$.

t	Y	$Y' = \log_{10}(Y)$	t^2	tY'
-1	4	0.60206	1	-0.60206
0	12	1.07918	0	0.00000
1	36	1.55630	1	1.55630
$\Sigma t=0$	-	$\Sigma Y'=3.23754$	$\Sigma t^2=2$	$\Sigma tY'=0.95424$

Calculate our transformed parameters a' and b' :

$$a' = \Sigma Y' / n = 3.23754 / 3 = 1.07918$$

$$b' = \Sigma tY' / \Sigma t^2 = 0.95424 / 2 = 0.47712$$

To convert these back to our original units, take the antilogarithm (base 10) of both parameters:

$$a = 10^{1.07918} = 12.0$$

$$b = 10^{0.47712} = 3.0$$

This gives us our final exponential trend curve equation:

$$Y_t = 12.0 \times (3.0)^t$$

Interpretation: The baseline revenue at our origin ($t = 0$) is \$12 million. The growth factor of $b = 3.0$ confirms that revenue is tripling each period, reflecting explosive exponential expansion.

PROBLEM 30: DE-SEASONALIZING SALES DATA FOR TREND EVALUATION

A company's actual observed sales revenue in Q4 is \$240,000. Their historical seasonal index for the fourth quarter is 120. Calculate the de-seasonalized sales volume to evaluate the true underlying trend.

SOLUTION

Seasonal fluctuations can obscure the true performance of a business. To remove these effects and see the underlying trend, we calculate the **de-seasonalized sales** by dividing the observed sales by the seasonal index (expressed as a decimal):

$$\text{De-seasonalized Sales} = [\text{Observed Sales} / \text{Seasonal Index}] \times 100$$

Substitute our known values into the formula:

$$\text{De-seasonalized Sales} = [240,000 / 120] \times 100 = 2,000 \times 100 = 200,000$$

Interpretation: While the company recorded \$240,000 in actual sales, our analysis reveals that \$40,000 of that volume was driven entirely by the predictable holiday sales surge. The true baseline trend value for this period is \$200,000. Comparing this de-seasonalized value against previous quarters allows management to accurately evaluate whether the business is growing or shrinking.